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By Amit Katiyar (MCA-JNU)



MATHEMATICS

- If $|k| = 5$ and $0^\circ \leq \theta \leq 360^\circ$, then the number of distinct solutions of $3 \cos \theta + 4 \sin \theta = k$ is
(a) 0 (b) 1
(c) 2 (d) infinite
- The area of the region bounded by the X-axis and the curves defined by $y = \tan x, -\frac{\pi}{3} \leq x \leq \frac{\pi}{3}$ and $y = \cot x, \frac{\pi}{6} \leq x \leq \frac{3\pi}{2}$ is
(a) $-\frac{1}{2} \log 2$ (b) $\frac{1}{2} \log 2$
(c) $\log 2$ (d) None of these
- If $\log(1 - x + x^2) = a_1x + a_2x^2 + a_3x^3 + \dots$ Then, $a_3 + a_6 + a_9 + \dots$ is equal to
(a) $\log 2$ (b) $\frac{2}{3} \log 2$
(c) $\frac{1}{3} \log 2$ (d) $2 \log 2$
- $\lim_{x \rightarrow \infty} \left(\frac{x+7}{x+2}\right)^{x+5}$ is equal to
(a) e^5 (b) e^{-5}
(c) e^2 (d) e^{-2}
- The number of common tangents to the circle $x^2 + y^2 = 4$ and $x^2 + y^2 - 6x - 8y = 24$ is
(a) 0 (b) 1
(c) 3 (d) 4
- In a ΔABC , if $\tan^2 \frac{A}{2} + \tan^2 \frac{B}{2} + \tan^2 \frac{C}{2} = k$, then k is always
(a) > 2 (b) ≥ 1
(c) $= 2$ (d) $= 1$
- Let $a = 2\hat{i} + \hat{j} + 2\hat{k}$, $b = \hat{i} - \hat{j} + 2\hat{k}$ and $c = \hat{i} + \hat{j} - 2\hat{k}$ are three vectors. Then, a vector in the plane of a and c whose projection on b is of magnitude $\frac{1}{\sqrt{6}}$ is
(a) $3\hat{i} - 2\hat{j}$ (b) $3\hat{i} + 2\hat{j}$
(c) $2\hat{i} + 3\hat{j} - \hat{k}$ (d) $3\hat{i} + 2\hat{j} + \hat{k}$
- For what value of p, the polynomial $x^4 - 3x^3 + 2px^2 - 6$ is exactly divisible by $(x-1)$?
(a) 2 (b) 4 (c) 6 (d) 8
- If $F(\theta) = \begin{bmatrix} \cos \theta & -\sin \theta & 0 \\ \sin \theta & \cos \theta & 0 \\ 0 & 0 & 1 \end{bmatrix}$, then $F(\theta)F(\alpha)$ is equal to
(a) $F(\theta\alpha)$ (b) $F\left(\frac{\theta}{\alpha}\right)$ (c) $F(\theta + \alpha)$ (d) $F(\theta - \alpha)$
- If $\frac{n!}{2!(n-2)!}$ and $\frac{n!}{4!(n-4)!}$ are in the ratio 2:1, then the value of n is
(a) 0 (b) 2 (c) 4 (d) 5
- The locus of the point of intersection of tangents to the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ which meet right angles is
(a) a circle (b) a parabola
(c) an ellipse (d) a hyperbola
- If the position vector of A and B relative to O be $\hat{i} - 4\hat{j} + 3\hat{k}$ and $-\hat{i} + 2\hat{j} - \hat{k}$ respectively, then the median through O of ΔABC is
(a) $-2\hat{i} + 2\hat{k}$ (b) $-\hat{j} + \hat{k}$
(c) $-\hat{i} - \hat{j} + \hat{k}$ (d) $-\hat{i} - \hat{j} - \hat{k}$
- The general value of θ , satisfying the equation, $\sin \theta = -\frac{1}{2}$, $\tan \theta = \frac{1}{\sqrt{3}}$ is
(a) $n\pi + \frac{\pi}{6}, n \in I$ (b) $n\pi + (-1)^n \left(\frac{7\pi}{6}\right), n \in I$
(c) $2n\pi \pm \frac{7\pi}{6}, n \in I$ (d) $2n\pi + \frac{11\pi}{6}, n \in I$

- The area of the triangle formed by the vertices whose position vectors are $3\hat{i} + \hat{j}, 5\hat{i} + 2\hat{j} + \hat{k}, \hat{i} - 2\hat{j} + 3\hat{k}$ is
(a) $\sqrt{21}$ sq. units (b) $\sqrt{23}$ sq. units
(c) $\sqrt{33}$ sq. units (d) $\sqrt{29}$ sq. units
- The standard deviation of 20 number is 30. If each of the numbers is increased by 4, then the new standard deviation will be
(a) 24 (b) 34 (c) 30 (d) 20
- The function $f(x) = \frac{x}{1+x \tan x}, 0 \leq x \leq \frac{\pi}{2}$ is maximum when
(a) $x = \sec x$ (b) $x = \tan x$
(c) $x = \cos x$ (d) None of the above
- If $f: R \rightarrow R$ is defined by
$$f(x) = \begin{cases} \frac{x+2}{x^2+3x+2}, & \text{if } x \in R - \{-1, -2\} \\ -1, & \text{if } x = -2 \\ 0, & \text{if } x = -1 \end{cases}$$
Then, $f(x)$ is continuous on the set
(a) R (b) $R - \{-2\}$
(c) $R - \{-1\}$ (d) $R - \{-1, -2\}$
- A polygon has 44 diagonals, the number of sides are
(a) 9 (b) 10 (c) 11 (d) 12
- The probability of occurrence of two events E and F are 0.25 and 0.50, respectively. The probability of their simultaneous occurrence is 0.14. The probability that neither E nor F occur is
(a) 0.61 (b) 0.11 (c) 0.39 (d) 0.89
- If $y = \tan^{-1} \left(\frac{3x-x^3}{1-3x^3} \right), -\frac{1}{\sqrt{3}} < x < \frac{1}{\sqrt{3}}$, then $\frac{dy}{dx}$ is
(a) $-\frac{1}{1+x^2}$ (b) $\frac{3}{1+x^2}$ (c) $\frac{3}{\sqrt{1+x^2}}$ (d) $\frac{1}{\sqrt{1+x^2}}$
- The value of $\tan 9^\circ - \tan 27^\circ - \tan 63^\circ + \tan 81^\circ$ is equal to
(a) 5 (b) 3 (c) 4 (d) 6
- In a triangle, if the sum of two sides is x and their product is y such that $(x+z)(x-z) = y$, where z is the third side of the triangle, then triangle is
(a) equilateral (b) right angled
(c) isosceles (d) obtuse angled
- If $H_1, H_2, \dots, H_n$ are n harmonic means between a and b ($\neq a$), then $\frac{H_1+a}{H_1-a} + \frac{H_n+b}{H_n-b}$
(a) 2n (b) n+1 (c) n-1 (d) 2n+1
- If $y = \sin^{-1} \left(\frac{x^2+1}{\sqrt{1+3x^2+x^4}} \right) (x > 0)$, then $\frac{dy}{dx} =$
(a) $\frac{x^2-1}{x^4+3x^2+1}$ (b) $\frac{x^2+1}{x^4+3x^2+1}$
(c) $\frac{x^2-1}{x^4-3x^2+1}$ (d) $\frac{x^2+1}{x^4-3x^2+1}$
- If $32 \tan^8 \theta = 2 \cos^2 \alpha - 3 \cos \alpha$ and $3 \cos 2\theta = 1$, then the general value of α for
(a) $n\pi \pm \frac{\pi}{3}$ (b) $2n\pi \pm \frac{2\pi}{3}$
(c) $2n\pi \pm \frac{\pi}{3}$ (d) $n\pi \pm \frac{2\pi}{3}$
- In a ΔABC , $\alpha \cos^2 \frac{C}{2} + c \cos^2 \frac{A}{2} = \frac{3b}{2}$, then the sides of the triangle are in
(a) AP (b) GP
(c) HP (d) None of the above
- If θ is acute between the pair of lines $x^7 - 7xy + 12y^2 = 0$, then $\frac{2 \cos \theta + 3 \sin \theta}{4 \sin \theta + 5 \cos \theta} =$
(a) $\frac{29}{69}$ (b) $\frac{61}{39}$
(c) $\frac{39}{61}$ (d) $\frac{69}{29}$



28. The four geometric means between 2 and 64 are
(a) $\frac{1}{4}, \frac{1}{8}, \frac{1}{16}, \frac{1}{32}$ (b) 4, 8, 16, 32
(c) $4\sqrt{2}, 8, 16\sqrt{2}, 32$ (d) None of the above
29. The lines $px + qy = 1$ and $qx + py = 1$ are respectively the sides AB, AC of the ΔABC and the base BC is bisected at (p, q) . Equation of the median of the triangle through the vertex A is
(a) $(2pq-1)(qx+py-1)-(p^2+q^2-1)(px+qy-1)=0$
(b) $(2pq-1)(px+qy-1)+(p^2+q^2-1)(qx+py-1)=0$
(c) $(2pq-1)(px+qy-1)-(p^2+q^2-1)(qx+py-1)=0$
(d) $(2pq-1)(qx+qy-1)+(p^2+q^2-1)(px+qy-1)=0$
30. If the vector $a\hat{i} + \hat{j} + \hat{k}, \hat{i} + b\hat{j} + \hat{k}, \hat{i} + \hat{j} + c\hat{k} (a, b, c \neq 1)$ are coplanar, then $\frac{1}{1-a} + \frac{1}{1-b} + \frac{1}{1-c} =$
(a) 0 (b) 1 (c) 2 (d) 3
31. Let $a = \hat{i} + \hat{j}$ and $b = 2\hat{i} - \hat{k}$. Then, the point of intersection of the lines $r \times a = b \times a$ and $r \times b = a \times b$ is
(a) $-\hat{i} + \hat{j} + \hat{k}$ (b) $3\hat{i} - \hat{j} + \hat{k}$
(c) $\hat{i} - \hat{j} - \hat{k}$ (d) $3\hat{i} + \hat{j} - \hat{k}$
32. If the system of equations $3x - y + 4z = 3, x + 2y - 3z = -2, 6x + 5y + \lambda z = -3$ has at least one solution, then λ is equal to
(a) -5 (b) 3 (c) 5 (d) 6
33. If a variable takes value 0, 1, 2, ..., 50 with frequencies $1, {}^{50}C_1, {}^{50}C_2, \dots, {}^{50}C_{50}$, then AM is
(a) 50 (b) 25 (c) $\frac{2^{50}}{50}$ (d) 51
34. If $A = \{1, 2, 3, 4\}$ and $B = \{3, 4, 5\}$, then the number of elements in $(A \cup B) \times (A \cap B) \times (A \Delta B)$
(a) 18 (b) 20 (c) 24 (d) 30
35. If n is an integer between 0 to 21, then find a value of n for which the value of $n!(21-n)!$ is minimum.
(a) 9 (b) 10 (c) 12 (d) 21
36. Suppose $A_1, A_2, \dots, A_{30}$ are 30 sets (each with five elements) and $B_1, B_2, B_3, \dots, B_n$ are n sets (each with three elements) such that $\bigcup_{i=1}^{30} A_i = \bigcup_{j=1}^n B_j = S$ and each element of S belong to exactly ten of the A_i 's and exactly 9 of the B_j 's. Then, n is equal to
(a) 15 (b) 45 (c) 75 (d) 90
37. The eccentric angle of the extremities of latusrectum of the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ are given by
(a) $\tan^{-1} \left(\pm \frac{ae}{b} \right)$ (b) $\tan^{-1} \left(\pm \frac{be}{e} \right)$
(c) $\tan^{-1} \left(\pm \frac{b}{ae} \right)$ (d) $\tan^{-1} \left(\pm \frac{a}{be} \right)$
38. If $\alpha \neq \beta$ and $a^2 = 5\alpha - 3, \beta^2 = 5\beta - 3$, then the equation whose roots are $\frac{\alpha}{\beta}$ and $\frac{\beta}{\alpha}$ is
(a) $3x^2 - 25x - 3 = 0$ (b) $3x^2 + 5x + 3 = 0$
(c) $3x^2 - 5x + 3 = 0$ (d) $3x^2 - 19x + 3 = 0$
39. The probability that a man who is x years old will die in a year is p . Then, amongst n persons $A_1, A_2, \dots, A_n$ each x year old now, the probability that A_1 will die in one year is
(a) $\frac{1}{n^2}$ (b) $1 - (1-p)^n$
(c) $\frac{1}{n^2} [1 - (1-p)^n]$ (d) $\frac{1}{n} [1 - (1-p)^n]$
40. Angle between a and b is 120° . If $|b| = 2|a|$ and the vectors $a + xb, a - b$ are at right angle, then x is equal to
(a) $\frac{1}{3}$ (b) $\frac{1}{5}$ (c) $\frac{2}{3}$ (d) $\frac{2}{5}$
41. If a number x is selected at random from natural numbers 1, 2, ..., 100, then the probability for $x + \frac{100}{x} > 29$ is
(a) $\frac{37}{50}$ (b) $\frac{39}{50}$ (c) $\frac{41}{50}$ (d) $\frac{43}{50}$
42. If X and Y are two sets, then $X \cap Y' \cap (X \cup Y)'$ is
(a) X' (b) Y'
(c) $\emptyset$ (d) None of these
43. $\int e^x (\sinh x + \cosh x) dx$ is equal to
(a) $e^x \sec h x + C$ (b) $e^x \cosh x + C$
(c) $\sinh 2x + C$ (d) $\cosh 2x + C$
44. If $e_1 = (1, 1, 1)$ and $e_2 = (1, 1, -1)$ and a and b are two vectors such that $e_1 = 2a + b$ and $e_2 = a + 2b$, then angle between a and b is
(a) $\cos^{-1} \left(-\frac{7}{11} \right)$ (b) $\cos^{-1} \left(\frac{7}{11} \right)$
(c) $\cos^{-1} \left(\frac{7}{9} \right)$ (d) $\cos^{-1} \left(\frac{6\sqrt{2}}{11} \right)$
45. If $P(1, 2), Q(4, 6), R(5, 7)$ and $S(a, b)$ are the vertices of a parallelogram PQRS, then
(a) $a=2, b=3$ (b) $a=3, b=4$
(c) $a=2, b=4$ (d) $a=3, b=5$
46. If $\alpha \cos \theta + b \sin \theta = 2$ and $\alpha \sin \theta - b \cos \theta = 3$, then $a^2 + b^2$ is equal to
(a) 6 (b) 5 (c) 13 (d) 10
47. The three thrown of three dice, the probability of throwing triplets not more than twice is
(a) $1 - \frac{1}{6^2}$ (b) $1 - \frac{1}{6^3}$ (c) $1 - \frac{1}{36^2}$ (d) $1 - \frac{1}{36^3}$
48. $\int 3^{3^{3^x}} \cdot 3^{3^x} \cdot 3^x dx$ is equal to
(a) $\frac{3^{3^{3^x}} \cdot 3^x}{(\log 3)^3} + C$ (b) $\frac{3^x}{(\log 3)^3} + C$
(c) $\frac{3^{3^x}}{(\log 3)^3} + C$ (d) $\frac{3^{3^{3^x}}}{(\log 3)^3} + C$
49. There are 50 questions in a paper. Find the number of ways in which a student can attempt one or more questions.
(a) $2^{50} - 1$ (b) $2^{50} + 1$
(c) $2^{50} - 2$ (d) $2^{50} + 2$
50. Consider the following frequency distribution table.
- | Class Interval | 10-20 | 20-30 | 30-40 | 40-50 | 50-60 | 60-70 | 70-80 |
|----------------|-------|-------|-------|-------|-------|-------|-------|
| Frequency | 180 | f_1 | 34 | 180 | 136 | f_2 | 50 |
- If the total frequency is 686 and the median is 42.6, then the value of f_1 and f_2 are
(a) 81, 25 (b) 82, 24
(c) 83, 23 (d) 84, 22

LOGICAL REASONING

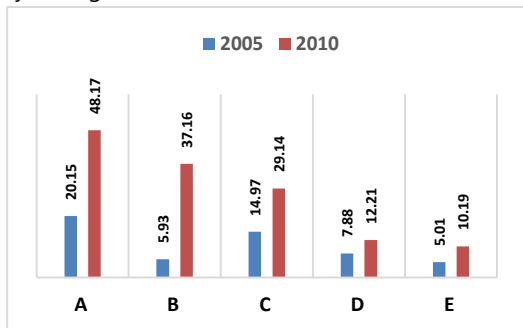
51. Choose the word which has the same relationship among the given three words: Yellow, Blue, Red
(a) Black (b) Orange
(c) White (d) Paint





52. A company produces five different products. The sales of these five products (in lakh number of packs) during 2005 and 2010 are shown in the following bar- graph. Question is based on this graph.

Sales (in lakh number of packs) of five different products of a company during 2005-2010.



PRODUCT

The sales of product A in 2010 was by that percent more than the sales of product B in 2010? (rounded off to the nearest integer)

- (a)33% (b)31% (c)28% (d)22%
53. Choose the pair of number that comes next:
1 10 7 20 13 30 19
- (a)26, 24 (b)25, 23 (c) 40, 25 (d)25, 22
54. Read the information given below and answer the following question:
- (i) In a family of six persons A, B, C, D, E and F, there are two married couples.
- (ii) D is grandmother of A and mother of B.
- (iii) C is wife of B and mother of F.
- (iv) F is the granddaughter of E.
- Who among the following is one of the couples?
- (a)CD (b)DE (c)EB (d)None of these
55. If $Z = 52$ and $ACT = 48$, then BAT will be equal to
- (a)46 (b)39 (c)44 (d)41

Directions (Q. Nos 56 and 57) Read the information given below and answer the following question.

- (i) In a family of six persons A, B, C, D, E and F there are two married couples.
- (ii) D is the grandmother of A and mother of B.
- (iii) C is wife of B and mother of F
- (iv) F is the granddaughter of E.
56. Which of the following is true?
- (a) A is brother of F. (b) A is sister of F
- (c) D has two grandsons (d) None of these
57. What is C to A?
- (a)Daughter (b)grandmother
- (c) mother (d) Cannot be determined
58. Find the matching term for Music: Guitar: Performer.
- (a) Dance: Tune: Instrument (b) Food: Recipe: Cook
- (c) Patient: Medicine :doctor (d) Trick : Rope : Acrobat

59. If $9 \times 3 + 8 = 24$, $10 \times 2 + 7 = 35$ and $80 \times 40 + 3 = 6$, then find the value of $12 \times 4 + 3 = ?$
- (a)7 (b)9 (c)12 (d)16

60. Choose the missing term out of the given alternatives: Q1F, S2E, U6D, W21C, ?
- (a)Y66B (b)Y44B (c)Y88B (d)Z88C

61. Study the following table carefully and answer the following question:

Subjects (full Marks 50)	40 and above	30 and above	20 and above	10 and above	0 and above
Physics	9	32	80	92	100
Chemistry	4	21	66	81	100
Aggregate	7	27	73	87	100

If it is known that atleast 23 students were eligible for a Symposium on Chemistry the minimum qualifying marks in Chemistry for eligibility to Symposium would lie in the range:

- (a)30-40 (b)20-30
- (c)Below 20 (d)Cannot be determined
62. In a certain code language, "do re me" means " he is late"; "fa me la" means " she is early" and "so ti do" means " he leaves soon". Which word in the language means 'late' ?
- (a)la (b)do (c)me (d)re
63. Find the missing term 7:56::9:?.
- (a)63 (b)81 (c) 90 (d)99
64. In this series, looking at the letter pattern fill the blank in the middle of the series:
ELFA, GLHA, ILJA,, MLNA
- (a)OLPA (b)KLMA (c)LLMA (d)KLLA
65. Read the following information carefully and then answer the following question:
- (i) $P \Psi Q$ means P is mother of Q
- (ii) $P \in Q$ means P is sister of Q
- (iii) $P \$ Q$ means P is father of Q
- (iv) $P \# Q$ means P is brother of Q
- Which of the following means R is brother of T?
- (a) $R \Psi S \# U \$ T$ (b) $U \Psi R \# S \# T$
- (c) $U \Psi R \in S \Psi T$ (d) $T \# S \$ Q \in R$
66. Choose the pair that best represents a similar relationship to the one expressed in the original pair of words:
WAITRESS : RESTAURANT
- (a) doctor: diagnosis (b) actor: role
- (c) driver : truck (d) teacher : school
67. COUNSEL is to BITIRAK, so also GUIDANCE is to?
- (a)EOHYZKBB (b)FOHYZJBB
- (c)FPHZZKAB (d) HOHYBJBA
68. Choose the number pair or group that is different from other?
- (a)15:46 (b)12:37
- (c)9:28 (d)8:33
69. If it is Saturday on January 1, 2000, then January 1, 2001 would have been
- (a)Monday (b)Sunday
- (c)Tuesday (d)Friday



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70. Radha is twice as old as Rita was 2 yr ago. If difference between their ages is 2 yr, how old is Radha today?

- (a)6 (b)8 (c)10 (d)12

71. How many times in a day are the hands of a clock pointing opposite to each other?

- (a)4 (b)20 (c)22 (d)14

Directions (Q.72 and 73) Studying the following table carefully and answer the following questions.

Subjects (full Marks 50)	40 and above	30 and above	20 and above	10 and above	0 and above
Physics	9	32	80	92	100
Chemistry	4	21	66	81	100
Aggregate	7	27	73	87	100

72. If atleast 60% marks in physics are required for pursuing higher studies in Physics. How many students will be eligible to pursue higher studies in Physics?

- (a)27 (b)32 (c)34 (d)41

73. The number of students scoring less than 40% marks in aggregate is

- (a)13 (b)19 (c)27 (d)20

74. Choose the missing term: BYCX : DWEV :: FUGT:?

- (a)EHJI (b)GHIJ (c)HSIR (d)SRHS

75. Read the information given below and answer the following question:

- (i) In a family of six persons A,B,C,D,E and F, there are two married couples.
- (ii) D is grandmother of A and mother of B.
- (iii) C is wife of B and mother of F
- (iv) F is the granddaughter of E

How many male members are there in the family?

- (a)Three (b)Two
(c)cannot be determined (d)None of these

76. Read the following information carefully and then answer the following question:

- (i) $P \Psi Q$ means P is mother of Q
- (ii) $P \in Q$ means P is sister of Q
- (iii) $P \$ Q$ means P is father of Q
- (iv) $P \# Q$ means P is brother of Q

If $K \Psi L \in M \# N$, then how K is related with N?

- (a)Mother (b)Aunt
(c)Grandfather (d)Grandmother

77. Choose the correct option for the remainder when $X = 1! + 2! + 3! + \dots + 100!$ is divided by 24.

- (a)9 (b)11 (c)152 (d)13

78. There are six members in a family. A is the father of D,E is the grandfather of D. B is the daughter-in-law of C.F is the uncle of D. What is the relationship of C with F?

- (a) Sister (b)Mother-in-law
(c) Nephew (d)Data inadequate

79. Read the following information carefully and then answer the following question:

(i) $P \Psi Q$ means P is mother of Q

(ii) $P \in Q$ means P is sister of Q

(iii) $P \$ Q$ means P is father of Q

(iv) $P \# Q$ means P is brother of Q

Which of the following means N is definitely daughter of K?

- (a) $K \$ L \# M \# N$ (b) $M \Psi K \$ N \in L$
(c) $K \$ M \# L \in N$ (d) $L \Psi K \$ N \# M$

80. Choose the word that is a necessary part of the underlined word:

HARVEST

- (a)autumn (b)crop (c)stockpile (d)tractor

81. Study the following table carefully and answer the following question.

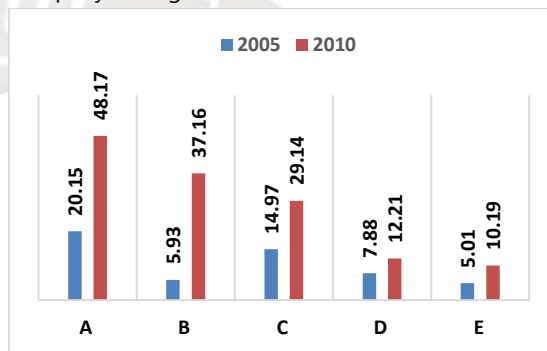
Subjects (full Marks 50)	40 and above	30 and above	20 and above	10 and above	0 and above
Physics	9	32	80	92	100
Chemistry	4	21	66	81	100
Aggregate	7	27	73	87	100

What is the different between the number of students passed with 30 as cut- off marks in Chemistry and those passed with 30 as cut- off marks in aggregate?

- (a)3 (b)4 (c)5 (d)6

82. A company produces five different products. The sales of these five products (in lakh number of packs) during 2005 and 2010 are shown in the following bar-graph. Question is based on this graph.

Sales (in lakh number of packs) of five different products of a company during 2005-2010.



PRODUCT

During the period 2005 - 2010. The minimum rate of increase in sales is in the case of

- (a)Product A (b)Product B
(c)Product D (d)Product E

83. Insert the missing number 8, 7, 11, 12, 14, 17, 17, 22,?

- (a)27 (b)20 (c)22 (d)24

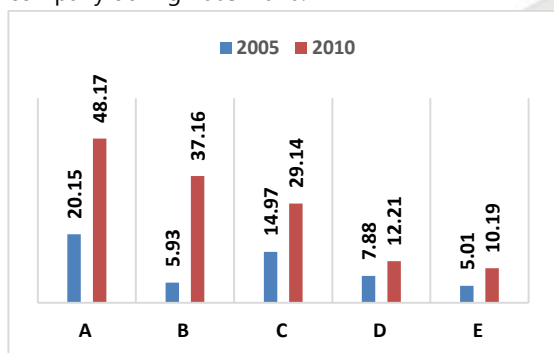
84. Choose the pair that best represents a similar relationship to be one expressed in the original

Pair of words: **PULSATE : THROB**

- (a) walk : run (b) tired: sleep
(c) examine : scrutinize (d) ballet: dancer



85. In a class of 50 students, Raghu's rank is twice that of Paul. There are 10 students who have ranked worse than that of Raghu. Paul's rank in the class is
(a) 5th (b) 10th (c) 15th (d) 20th
86. Choose the odd one:
(a) Zebra (b) Rhinoceros (c) Hyena (d) Giraffe
87. If 'A3T15R' stands for 'ACTOR' and 'D1T5' stands for DATE. How will you code 'ROTATE'?
(a) R16T1T5 (b) R15T1T5 (c) R15T1T6 (d) R15C1T7
88. A company produces five different products. The sales of these five products (in lakh number of packs) during 2005 and 2010 are shown in the following bar-graph. Question is based on this graph.
Sales (in lakh number of packs) of five different products of a company during 2005-2010.



PRODUCT

The sales have increased by nearly 55% from 2005 to 2010 in case of

- (a) Product A (b) Product B
(c) Product C (d) Product D
89. Which word does NOT belong with the others?
(a) wing (b) fin (c) break (d) rudder
90. Insert the missing number: 16, 33, 65, 131, 261, ?
(a) 523 (b) 521 (c) 613 (d) 721

COMPUTER

91. The process when processor fetch or decode another instruction during the execution of current instruction is called
(a) Supercomputing (b) Pipelining
(c) Cloud computing (d) Grid computing
92. Which of the following is used by ALU to store the intermediate results?
(a) Stack (b) Heap
(c) Registers (d) Accumulators
93. One TeraByte (TB) = GB and One EXAByte (EB) = GB
(a) $2^{10}GB$, $2^{16}GB$ (b) $2^{10}GB$, $2^{20}GB$
(c) $2^{10}GB$, $2^{24}GB$ (d) $2^{10}GB$, $2^{30}GB$
94. The cache memory is more effective because of
(a) Memory localization
(b) Locality of reference
(c) Memory size
(d) None of the mentioned

95. Which of the following is the fastest means of memory access for CPU?
(a) Registers (b) cache
(c) Main memory (d) Stack
96. The number $(2217)_8$ is equivalent to
(a) $(608)_{16}$ (b) $(028F)_{16}$ (c) $(048F)_{16}$ (d) $(2297)_{10}$
97. To fetch data from the secondary memory which one of the following register is used?
(a) MAR (b) PC (c) IR (d) MBR
98. The binary multiplication 00×11 will give
(a) 11 (b) 00 (c) 01 (d) 10
99. Consider a computer system with speed of 10^6 instructions per second. A program P, having $2n^2$ steps is run on this system, where n is the input size. If $n = 10000$, what is the execution time for P?
(a) 1.2 (b) 20s (c) 100s (d) 200s
100. To access the I/O devices, the status flag is continuously checked in
(a) Program controlled I/O (b) Memory mapped I/O
(c) I/O mapped (d) None of these

GENERAL ENGLISH

Directions (Q. Nos.101-102) Read the following passage carefully and answer the question that follow: It is said with truth that the function of a university is to prepare the young to take their place in human society. It must provide its members with the knowledge and skill necessary to make them efficient citizens. But is the whole duty of man exhausted by the acquisition of knowledge and professional training? Is a university only an institution for higher learning, a factory which churns out clerks and technicians able to run the machinery of the State? Mere knowledge which gratifies curiosity is different from culture which refines personality. Culture is not remembering a mass of serious details about the dates of birth of the great heroes of the world or the interesting names of the fastest ships which cross the Atlantic or entertaining odds and ends gathered from the latest who's who. A well known institution of this country has for its motto sa vidya yavimuchyate: that is, knowledge which is designed for salvation, for the development of the soul, is the best. Such an idea is not merely an Indian idiosyncrasy. Plato said long ago that the culture of soul is "the first and fairest thing that the best of men can ever have. According to Goethe, the object of education is to form tastes and not simply to communicate knowledge. A man's culture is not judged by the amount of tabulated information which he has at his command, but by the quality of mind which he brings to bear on the facts of life. Education is not cramming the mind with a host of technical details, putting sight, as it were, into blind eyes.

The eye of the soul is never blind, only its gaze may be turned to the false and the fleeting. Too often the vision may be dragged downwards by the "lead weights" of pride and prejudice, of



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passion and desire. The function of the teacher is not to add to the "leaden weights" but remove them and liberate the soul from the encumbrance so that it may follow its native impulse to soar upwards. The student at a university does not merely learn something, but becomes something by being exposed, in the most elastic period of his life, to transforming influences, such as the constant clash of mind with mind, the interchange of ideas, the testing of opinions, and the growth of knowledge of human nature.

101. Mere knowledge and culture may be distinguished from each other in that :

- (a) the former widens the mental horizon, the letter enlarges the hearts
- (b) the former gratifies curiosity, the letter refines personality
- (c) the former is concerned with facts, the letter with fiction
- (d) the former adds to power, the letter adds to prestige

102. What is the function of education according to the ancient Indian philosophers?

- (a) Education is the development mind
- (b) Education brings about salvation and development of the soul
- (c) Education is the cultivation of culture
- (d) Education aims at the inculcation of generosity

103. Choose the most suitable synonym of the given word: SCINTILLATING

- (a) touching
- (b) nagging
- (c) glittering
- (d) warning

104. Choose the most suitable synonym of the given word : BLAND

- (a) pleasant
- (b) harsh
- (c) irritating
- (d) tasteless

Directions (Q. Nos.105-108) Read the following passage carefully and answer the question that follow: It is said with truth that the function of a university is to 1 prepare the young to take their place in human society. It must provide its members with the knowledge and skill necessary to make them efficient citizens. But is the whole duty of man exhausted by the acquisition of knowledge and professional training? Is a university only an institution for higher learning, a factory which churns out clerks and technicians able to run the machinery of the State? Mere knowledge which gratifies curiosity is different from culture which refines personality. Culture is not remembering a mass of serious details about the dates of birth of the great heroes of the world or the interesting names of the fastest ships which cross the Atlantic or entertaining odds and ends gathered from the latest who's who. A well known institution of this country has for its motto *avidyayavimuchyate*: that is, knowledge which is designed for salvation, for the development of the soul, is the best. Such an idea is not merely an Indian idiosyncrasy. Plato said long ago that the culture of soul is "the first and fairest thing that the best of men can ever have. According to Goethe, the object of education is to form tastes and not simply to communicate knowledge. A man's culture is not judged by the amount of tabulated information which he has at his command, but by the quality of mind which

he brings to bear on the facts of life. Education is not cramming the mind with a host of technical details, putting sight, as it were, into blind eyes. The eye of the soul is never blind, only its gaze may be turned to the false and the fleeting. Too often the vision may be dragged downwards by the "leaden weights" of pride and prejudice, of passion and desire. The function of the teacher is not to add to the "leaden weights" but remove them and liberate the soul from the encumbrance so that it may follow its native impulse to soar upwards. The student at a university does not merely learn something, but becomes something by being exposed, in the most elastic period of his life, to transforming influences, such as the constant clash of mind with mind, the interchange of ideas, the testing of opinions, and the growth of knowledge of human nature.

105. What is the meant by " leaden weights"?

- (a) Weights made of leaden
- (b) Cold scientific knowledge
- (c) Hindering factor, of pride and prejudice, passion, and desire, in the liberation of the soul
- (d) Social and family responsibilities

106. The function of university is

- (a) to enable the young to gather fact about the world
- (b) to learn to do his job
- (c) to prepare the young to take their place in society
- (d) to enable them to learn to talk with others

107. According to the passage, the function of the teacher is

- (a) to remove leaden weight of pride and prejudice, passion, and desire to liberate the soul
- (b) to instill facts into the minds of the students
- (c) to teach humanity
- (d) to foster brotherhood

108. What are the hindering factors in the liberation and development of the soul?

- (a) Power and wealth
- (b) Passion and desire, pride and prejudice
- (c) Money and influence
- (d) Greed and envy

109. Fill in the blank with the most appropriate word given:

I cannot.....it to you right now; tomorrow we will discuss about it:

- (a) demand
- (b) disturb
- (c) explain
- (d) expect

110. Choose the most suitable synonym of the given word?

RECUPERATE

- (a) recapture
- (b) reclaim
- (c) recover
- (d) recur

111. Fill in the blank with the most appropriate word:

The_____with which he is able to wield the paint brush is really remarkable?

- (a) ease
- (b) practice
- (c) sweep
- (d) gait

Directions (Q. Nos. 112-113) Read the following passage carefully and answer the questions that follow:





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It is said with truth that the function of a university is to prepare the young to take their place in human society. It must provide its members with the knowledge and skill necessary to make them efficient citizens. But is the whole duty of man exhausted by the acquisition of knowledge and professional training? Is a university only an institution for higher learning, a factory which churns out clerks and technicians able to run the machinery of the State? Mere knowledge which gratifies curiosity is different from culture which refines personality. Culture is not remembering a mass of serious details about the dates of birth of the great heroes of the world or the interesting names of the fastest ships which cross the Atlantic or entertaining odds and ends gathered from the latest who's who. A well known institution of this country has for its motto *avidyayavimuchyate*: that is, knowledge which is designed for salvation, for the development of the soul, is he best. Such an idea is not merely an Indian Hiosyncrasy. Plato said long ago that the culture of soul is the first and fairest thing that the best of men can ever have. According to Goethe, the object of education is to form tastes and not simply to communicate knowledge. A man's culture is not judged by the amount of tabulated information which he has at his command, but by the quality of mind which he brings to bear on the facts of life. Education is not cramming the mind with a host of technical details, putting sight, as it were, into blind eyes. The eye of the soul is never blind, only its gaze may be turned to the false and the fleeting. Too often the vision may be dragged downwards by the "leaden weights" of pride and prejudice, of passion and desire. The function of the teacher is not to add to the "leaden weights" but remove them and liberate the soul from the encumbrance so that it may follow its native impulse to soar upwards. The student at a university does not merely learn something, but becomes something by being exposed, in the most elastic period of his life, to transforming influences, such as the constant clash of mind with mind, the interchange of ideas, the testing of opinions, and 'the growth of knowledge of human nature.

112. What is the object of education according to Goethe?"

- (a) It teaches social manners
- (b) It teaches courtesy
- (c) It communicates knowledge
- (d) It forms taste

113. What is the man's culture judged by?

- (a) By the quality of mind which he brings to bear on the facts of life
- (b) By man's social skill
- (c) by the variety of books he reads
- (d) By money and influence

114. Choose the most suitable antonym of given word : ERUDITE

- (a) scholarly
- (b) unfamiliar
- (c) illiterate
- (d) arrogant

115. Choose one word for the following

An uncivilized/primitive person.

- (a) barbarian
- (b) ascetic
- (c) bourgeois
- (d) altruist

116. Fill in the blank with the most appropriate word. The car driver was arrested for rash driving and his license was..... by the police.

- (a) impounded
- (b) flown
- (c) penalized
- (d) banned

117. Fill in the blank with the most appropriate word:

The state Transportation Corporation has..... a loss of 5 crore INR this year.

- (a) derived
- (b) incurred
- (c) performed
- (d) formulated

118. Choose the most suitable antonym of given word. NEGLIGENCE

- (a) diligence
- (b) meticulousness
- (c) integrity
- (d) honest

119. Write one word for the following.

One who does not save for future.

- (a) incorporeal
- (b) inedible
- (c) improvident
- (d) implacable

120. Fill in the blank with the most appropriate word given in the options.

The child prefers room withceilings.

- (a) tall
- (b) long
- (c) higher
- (d) high





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ANSWER KEY

1. C	2. D	3. B	4. A	5. B	6. B	7. B	8. B
9. C	10. d	11.	12. b	13.	14. d	15. c	16. c
17. C	18. c	19. c	20. b	21. c	22. b	23. a	24. a
25. B	26. a	27. a	28. a	29. a	30. b	31. d	32. a
33. A	34. d	35. b	36. b	37. c	38. d	39. d	40. d
41. B	42. c	43. b	44. a	45. a	46. a	47. d	48. d
49. A	50. b	51. b	52. b	53. c	54. b	55. a	56. a
57. C	58. d	59. b	60. c	61. b	62. d	63. c	64. d
65. B	66. d	67. b	68. d	69. a	70. b	71. c	72. b
73. C	74. c	75. c	76. a	77. a	78. d	79. b	80. b
81. D	82. c	83. b	84. c	85. d	86. c	87. b	88. d
89. C	90. a	91. a	92. d	93. d	94. b	95. a	96. c
97. A	98. b	99. d	100. a	101. b	102. b	103. c	104. d
105. C	106. c	107. a	108. b	109. c	110. c	111. c	112. d
113. A	114. d	115. a	116. a	117. b	118. b	119. c	120. d

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MATHS SOLUTION

1. (c)

$$3\cos x + 4\sin x = 5 \text{ and } |K| = 5$$

$$\Rightarrow 3\cos x + 4\sin x = \pm 5$$

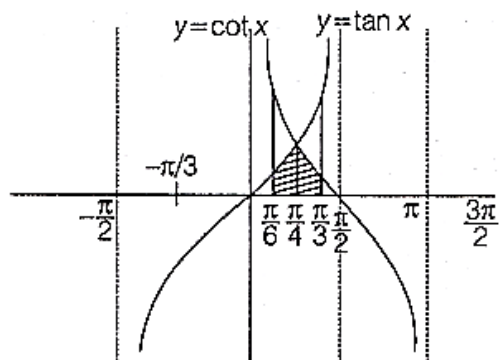
$$\Rightarrow K = \pm 5$$

$$\because -\sqrt{3^2 + 4^2} \leq 4\sin x + 3\cos x \leq \sqrt{3^2 + 4^2}$$

$$\Rightarrow -5 \leq 4\sin x + 3\cos x \leq 5$$

$$\Rightarrow 2 \text{ different solutions}$$

2. (d)



Area of shaded region

$$A = \int_{\pi/6}^{\pi/4} (\tan x) dx + \int_{\pi/4}^{\pi/3} (\cot x) dx$$

$$= (\log \sec x)_{\pi/6}^{\pi/4} + (\log \sin x)_{\pi/4}^{\pi/3}$$

$$= \left(\log \sqrt{2} - \log \frac{2}{\sqrt{3}} \right) + \left(\log \frac{\sqrt{3}}{2} - \log \frac{1}{\sqrt{2}} \right)$$

$$= \log \frac{3}{2}$$

3. (b)

$$\log(1 - x + x^2) = \left(x^3 - \frac{x^6}{2} + \frac{x^9}{3} - \frac{x^{12}}{4} + \dots \right) + \left(-x + \frac{x^2}{2} - \frac{x^3}{3} + \frac{x^4}{4} - \dots \right)$$

$$= -x + \frac{x^2}{2} + x^3 \left(1 - \frac{1}{3} \right) + \frac{x^4}{4} - \frac{x^5}{5} + x^6 \left(\frac{1}{2} + \frac{1}{6} \right) - \frac{x^7}{7} + \frac{x^8}{8} + \dots$$

$$= \frac{2}{3} + \left(\frac{-2}{6} \right) + \frac{2}{9} + \left(\frac{-2}{12} \right) + \dots$$

$$= \frac{2}{3} \left(1 - \frac{1}{2} + \frac{1}{3} - \frac{1}{4} + \dots \right) = \frac{2}{3} \log_2$$

4. (a)

$$\lim_{x \rightarrow \infty} \left(\frac{x+7}{x+2} \right)^{x+5} = \lim_{x \rightarrow \infty} \left(\frac{1+7/x}{1+2/x} \right)^{x+5} \text{ is an } 1^\infty \text{ form}$$

$$\lim_{x \rightarrow a} (f(x))^{g(x)} = e^{\lim_{x \rightarrow a} [f(x) - 1]g(x)}$$

$$= e^{\lim_{x \rightarrow \infty} \left(\frac{x+7}{x+2} - 1 \right) (x+5)}$$

$$= e^{\lim_{x \rightarrow \infty} \left(\frac{5}{x+2} \right) (x+5)}$$

$$= e^{5 \lim_{x \rightarrow \infty} \left(\frac{1+5/x}{1+2/x} \right)} = e^5$$

5. (b)

$$S_1: x^2 + y^2 = 4, S_2: x^2 + y^2 - 6x - 8y = 24$$

$$C_1(0,0) C_2(3,4)$$

$$r_1 = 2, r_2 = 7$$

$$\therefore C_1 C_2 = |r_1 - r_2| = 5$$

$\therefore$ Circle touches internally.

$\therefore$ Number of common tangents = 1

6. (b)

$$\frac{A+B}{2} = \frac{\pi}{2} - \frac{C}{2} \Rightarrow \tan \left(\frac{A+B}{2} \right) = \tan \left(\frac{\pi}{2} - \frac{C}{2} \right)$$

$$\Rightarrow \tan \frac{A}{2} \tan \frac{B}{2} + \tan \frac{B}{2} \tan \frac{C}{2} + \tan \frac{C}{2} \tan \frac{A}{2} = 1$$

$$\text{Let } \tan \frac{A}{2} = x, \tan \frac{B}{2} = y, \tan \frac{C}{2} = z$$

$$\text{Hence, } xy + yz + zx = 1$$

We should know

$$(x-y)^2 \geq 0 \Rightarrow x^2 + y^2 \geq 2xy$$

$$(y-z)^2 \geq 0 \Rightarrow y^2 + z^2 \geq 2yz$$

$$(z-x)^2 \geq 0 \Rightarrow z^2 + x^2 \geq 2zx$$

Add these three equations, we get

$$x^2 + y^2 + z^2 \geq xy + yz + zx$$

$$\text{Hence, } x^2 + y^2 + z^2 \geq 1$$

$$\Rightarrow \tan^2 \frac{A}{2} + \tan^2 \frac{B}{2} + \tan^2 \frac{C}{2} \geq 1$$

Alternate Solution

$$\text{For } A = B = C = \frac{\pi}{3}$$

$$= \tan^2 30^\circ + \tan^2 30^\circ + \tan^2 30^\circ = 3 \left(\frac{1}{\sqrt{3}} \right)^2 = 1$$

$$\text{For } A = 30^\circ, B = 60^\circ, C = 90^\circ$$

$$= \tan^2 15^\circ + \tan^2 30^\circ + \tan^2 45^\circ$$

$$= (2 - \sqrt{3})^2 + \frac{1}{3} + 1 = 4 + 3 - 4\sqrt{3} + \frac{4}{3} = 7 + \frac{4}{3} - 4\sqrt{3}$$

$$= \frac{25}{3} - 4\sqrt{3} = \frac{25}{3} - 4\sqrt{3} > 0$$

$$\text{So, } \tan^2 \frac{A}{2} + \tan^2 \frac{B}{2} + \tan^2 \frac{C}{2} \geq 1$$

7. (b)

$$\text{Let vector to } V = (2\hat{i} + \hat{j} + 2\hat{k}) + \lambda(\hat{i} + \hat{j} - 2\hat{k})$$

$$V = (2 + \lambda)\hat{i} + (1 + \lambda)\hat{j} + (2 - 2\lambda)\hat{k}$$

Using projection formula,

$$\frac{V \cdot b}{|b|} = \frac{1}{\sqrt{6}}$$

$$\Rightarrow \frac{2 + \lambda - 1 - \lambda + 4 - 4\lambda}{\sqrt{6}} = \frac{1}{\sqrt{6}}$$

$$\Rightarrow 5 - 4\lambda = 1 \Rightarrow 4\lambda = 4$$

$$\Rightarrow \lambda = 1$$

$$\Rightarrow v = 3\hat{i} + 2\hat{j}$$

8. (b) $\therefore$ divisible by $(x-1)$

$$\Rightarrow x-1 = 0 \text{ is factor of given polynomial}$$

$$\Rightarrow x = 1 \text{ satisfy}$$

$$\Rightarrow f(1) = (1)^4 - 3(1)^3 + 2(p)(1)^2 - 6 = 0$$

$$\Rightarrow 1 - 3 + 2p - 6 = 0$$

$$\Rightarrow p = 4$$



9. (c) $f(\alpha) = \begin{bmatrix} \cos \alpha & -\sin \alpha & 0 \\ \sin \alpha & \cos \alpha & 0 \\ 0 & 0 & 1 \end{bmatrix}$

$$f(\theta) \cdot f(\alpha) = \begin{bmatrix} \cos \theta & -\sin \theta & 0 \\ \sin \theta & \cos \theta & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} \cos \alpha & -\sin \alpha & 0 \\ \sin \alpha & \cos \alpha & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} \cos \theta \cos \alpha - \sin \theta \sin \alpha & -(\cos \theta \sin \alpha + \sin \theta \cos \alpha) & 0 \\ \sin \theta \cos \alpha + \sin \alpha \cdot \cos \theta & \cos \theta \cos \alpha - \sin \theta \sin \alpha & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$\begin{bmatrix} \cos(\theta + \alpha) & -\sin(\theta + \alpha) & 0 \\ \sin(\theta + \alpha) & \cos(\theta + \alpha) & 0 \\ 0 & 0 & 1 \end{bmatrix} = f(\theta + \alpha)$$

10. (d) Given

$$\frac{n!}{2!(n-2)(n-3)(n-4)!} = \frac{2}{1} \Rightarrow \frac{12}{n^2-5n+6} = 2$$

$$\Rightarrow n^2 - 5n = 0 \Rightarrow n = 5, n \neq 0$$

11. (a) Given that,

Ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ (i)

Equation of tangents on ellipse (i)

$$y = mx \pm \sqrt{a^2 m^2 + b^2}$$

$$\Rightarrow y^2 + m^2 x^2 = a^2 m^2 + b^2$$

$$\Rightarrow m^2 (x^2 - a^2) - 2mxy + y^2 - b^2 = 0$$

Which are perpendicular

$$\therefore m_1 m_2 = 1 \Rightarrow \frac{y^2 - b^2}{x^2 - a^2} = -1$$

$$\Rightarrow x^2 + y^2 = a^2 + b^2 \text{ which is an equation of circle.}$$

12. (b) Given that

$$OA = \hat{i} - 4\hat{j} + 3\hat{k} \text{ and } OB = -\hat{i} + 2\hat{j} - \hat{k}$$

Since, the median will pass through the mid-point of AB.

$$\text{Hence, median} = \frac{OA+OB}{2}$$

$$= \frac{(\hat{i} - 4\hat{j} + 3\hat{k}) + (-\hat{i} + 2\hat{j} - \hat{k})}{2} = -\hat{j} + \hat{k}$$

13. (c)

$$\sin \theta = -\frac{1}{2} = \sin \left(\pi + \frac{\pi}{6} \right) = \sin \frac{7\pi}{6}$$

$$\Rightarrow \theta = n\pi + (-1)^n \frac{7\pi}{6}, n \in \mathbb{I}$$

$$\tan \theta = \frac{1}{\sqrt{3}} = \tan \left(\pi + \frac{\pi}{6} \right) = \tan \frac{7\pi}{6}$$

$$\Rightarrow \theta = n\pi + \frac{7\pi}{6}; n \in \mathbb{I}$$

As we know, period of $\sin x$ is 2π and of $\tan x$ is π . LCM of their periods is 2π .

$$\text{Hence, answer is } \theta = 2n\pi + \frac{7\pi}{6}$$

14. (c) $AB = 2\hat{i} + \hat{j} + \hat{k}$ and $AC = -2\hat{i} - 3\hat{j} + 3\hat{k}$

$$\text{Area of triangle} = \frac{1}{2} |AB \times AC|$$

$$AB \times AC = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 2 & 1 & 1 \\ -2 & -3 & 3 \end{vmatrix}$$

$$= \hat{i}(3+3) - \hat{j}(6+2) + \hat{k}(-6+2) = 6\hat{i} - 8\hat{j} - 4\hat{k}$$

$$\text{Area} = |-3\hat{i} - 4\hat{j} - 2\hat{k}| = \sqrt{9+16+4} = \sqrt{29} \text{ units.}$$

15. (c)

Let the number $x_1, x_2, \dots, x_{20}$

Let the number $x_1, x_2, \dots, x_{20}$

$$\text{Mean}(\bar{x}) = \frac{x_1 + x_2 + \dots + x_{20}}{20}, \sigma(x) = \sqrt{\frac{\sum (x_i - \bar{x})^2}{n}} = 30$$

Let new numbers $y_1, y_2, y_3, \dots, y_{20}$

$$y_1 = x_1 + 4, y_2 = x_2 + 4,$$

$$y_3 = x_3 + 4 \dots, y_{20} = x_{20} + 4$$

$$\bar{y} = \frac{y_1 + y_2 + \dots + y_{20}}{20} = \bar{x} + 4$$

$$y_i - \bar{y} = (x_i + 4) - (\bar{x} + 4), \text{ where } i = 1, 2, 3 \dots 20$$

$$y_i - \bar{y} = x_i - \bar{x}$$

$$\sigma(y) = \sqrt{\frac{\sum (y_i - \bar{y})^2}{n}} = \sqrt{\frac{\sum (x_i - \bar{x})^2}{n}} = 30$$

Now, we know that if each term of data is either increased or decreased by the same constant number then the standard deviation of the data remains unchanged. And in question it's already given that the standard deviation of 20 numbers is 30.

16. (a)

$$f(x) = \frac{x}{1+x \tan x}, 0 \leq x \leq \frac{\pi}{2}$$

$$f'(x) = \frac{(1+x \tan x) - x(x \sec^2 x + \tan x)}{(1+\tan x)^2} = \frac{1-x^2 \sec^2 x}{(1+x \tan x)^2}$$

for minima / Maxima

$$f'(x) = 0 \Rightarrow 1 - x^2 \sec^2 x = 0$$

$$x^2 = \cos^2 x$$

$$x = \cos x$$

$$f''(x) > 0 \text{ at } x > \cos x$$

$$f''(x) < 0 \text{ at } x = \cos x$$

$$f(x) \text{ is maximum when } x = \cos x$$

17. (c)

If $f: R \rightarrow R$ is defined by

$$f(x) = \begin{cases} \frac{x+2}{x^2+3x+2}, & \text{if } x \in R - \{-1, -2\} \\ -1, & \text{if } x = -2 \\ 0, & \text{if } x = -1 \end{cases}$$

$\therefore f(x)$ is continuous for every value of R except $\{-1, -2\}$.

Now, we have to check continuity at $x = -2$ and $x = -1$

$$\lim_{x \rightarrow -2} f(x) = \lim_{x \rightarrow -2} \frac{x+2}{x^2+3x+2} = \frac{0}{0}$$

$$\lim_{x \rightarrow -2} \frac{(x+2)}{(x+1)(x+2)} = \frac{1}{-2+1} = -1 = f(-2)$$

$$\Rightarrow f(x) \text{ is continuous at } x = -2$$



Now at $x = -1$

$$\lim_{x \rightarrow -1} f(x) = \lim_{x \rightarrow -1} \frac{(x+2)}{(x+1)(x+2)} = \lim_{x \rightarrow -1} \frac{1}{x+1} = \infty$$

Limit does not exist.

$\Rightarrow f(x)$ is not continuous at $x = -1$.

- 18. (c)** $\because$ Number of diagonals in n -sided polygon is $(^nC_2 - n)$

$$\therefore {}^nC_2 - n = 44 \Rightarrow \frac{n(n-1)}{2} - n = 44$$

$$\Rightarrow n(n-3) = 88 \Rightarrow n^2 - 3n - 88 = 0$$

$$\Rightarrow (n-11)(n+8) = 0 \Rightarrow n=11, n \neq -8 \text{ (never be negative)}$$

- 19. (c)**

$$P(E) = 0.25, P(F) = 0.50$$

$$P(E \cap F) = 0.14$$

$$P(\bar{E} \cap \bar{F}) = P(\bar{E} \cup \bar{F})$$

$$= 1 - P(E \cup F) = 1 - P(E) - P(F) + P(E \cap F)$$

$$= 1 - 0.25 - 0.50 + 0.14 = 0.39$$

- 20. (b)** $y = \tan^{-1} \left(\frac{3x-x^3}{1-3x^2} \right)$

Let $x = \tan \theta$

$$\therefore y = \tan^{-1} \left(\frac{3 \tan \theta - \tan^3 \theta}{1 - 3 \tan^2 \theta} \right) = \tan^{-1} (\tan 3\theta) = 3\theta$$

$$\Rightarrow y = 3 \tan^{-1} x \quad [\theta = \tan^{-1} x]$$

$$\therefore \frac{dy}{dx} = 3 \frac{d}{dx} (\tan^{-1} x) = \frac{3}{1+x^2}$$

- 21. (c)**

$$\tan 9^\circ + \tan 81^\circ - (\tan 27^\circ + \tan 63^\circ)$$

$$\Rightarrow \tan 9^\circ + \cot 9^\circ - (\tan 27^\circ + \cot 27^\circ)$$

$$\Rightarrow \frac{\sin 9^\circ}{\cos 9^\circ} + \frac{\cos 9^\circ}{\sin 9^\circ} - \left(\frac{\sin 27^\circ}{\cos 27^\circ} + \frac{\cos 27^\circ}{\sin 27^\circ} \right)$$

$$\Rightarrow \frac{1}{\sin 9^\circ \cos 9^\circ} - \left(\frac{1}{\sin 27^\circ \cos 27^\circ} \right)$$

$$\Rightarrow \frac{2}{\sin 18^\circ} - \frac{2}{\sin 54^\circ} = \frac{2}{\sin 18^\circ} - \frac{2}{\cos 36^\circ}$$

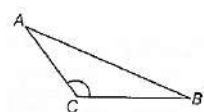
$$\Rightarrow 2 \left[\frac{4}{\sqrt{5}-1} - \frac{4}{\sqrt{5}+1} \right] = 8 \left[\frac{\sqrt{5}+1-\sqrt{5}+1}{5-1} \right] \Rightarrow \frac{16}{4} = 4$$

- 22. (d)** Let a and b be the sides of triangle

$$a + b = x \Rightarrow ab = y$$

$$\text{Given, } x^2 - z^2 = y$$

Here, z is its third side.



$$(a+b)^2 - z^2 = ab \Rightarrow a^2 + b^2 + 2ab - z^2 = ab$$

$$a^2 + b^2 + ab = z^2 \Rightarrow a^2 + b^2 - z^2 = -ab$$

$$\frac{a^2+b^2-z^2}{2ab} = -\frac{1}{2} \Rightarrow \cos C = \frac{2\pi}{3}$$

So, the triangle is obtuse angled.

- 23. (a)**

Given $H_1, H_2, \dots, H_6$ are harmonic means between a and b , ($a \neq b$)

$$\Rightarrow \frac{1}{H_1}, \frac{1}{H_2}, \frac{1}{H_3}, \dots, \frac{1}{H_n}$$

$$\Rightarrow \frac{1}{a} = \frac{dH_1}{a-H_1} \Rightarrow a = \frac{a-H_1}{dH_1}$$

$$\text{Also } b = \frac{b-H_n}{dH_n}$$

$$\text{Now, } \frac{H_1+a}{H_1-a} + \frac{H_n+b}{H_n-b} = \frac{1}{d} \left(\frac{1}{b} - \frac{1}{a} + \frac{1}{H_n} - \frac{1}{H_1} \right)$$

$$= (n+1) + (n-1) \left[\frac{1}{b} - \frac{1}{a} = d(n+1) \right]$$

$$= 2n.$$

- 24. (a)**

$$y = \sin^{-1} \left(\frac{x^2+1}{\sqrt{1+3x^2+x^4}} \right)$$

$$\text{So } \frac{dy}{dx} = \frac{1}{\sqrt{1-\frac{x^4+1+3x^2}{1+3x^2+x^4}}} \cdot \frac{d}{dx} \left(\frac{x^2+1}{\sqrt{1+3x^2+x^4}} \right) = \frac{\sqrt{1+3x^2+x^4}}{x}$$

$$\left\{ \frac{\sqrt{1+3x^2+x^4}(2x) - (x^2+1) \cdot \frac{1}{2\sqrt{1+3x^2+x^4}}(6x+4x^3)}{(1+3x^2+x^4)} \right\}$$

$$= \frac{\sqrt{1+3x^2+x^4}}{x(1+3x^2+x^4)} \left\{ \frac{2x(1+3x^2+x^4) - (x^2+1)(3x+2x^3)}{\sqrt{1+3x^2+x^4}} \right\}$$

$$= \frac{1}{x(1+3x^2+x^4)}$$

$$\{2x + 6x^3 + 2x^5 - 3x^3 - 2x^5 - 3x - 2x^3\}$$

$$= \frac{1}{x(1+3x^2+x^4)} \{4x^3 - x\}$$

$$\frac{dy}{dx} = \frac{x^2-1}{1+3x^2+x^4}$$

- 25. (b)**

$$32 \tan^8 \theta = 2 \cos^2 \alpha - 3 \cos \alpha \text{ and } 3 \cos 2\theta = 1$$

$$\Rightarrow \cos 2\theta = \frac{1}{3} \Rightarrow 2 \cos^2 \theta - 1 = \frac{1}{3}$$

$$\Rightarrow 2 \cos^2 \theta = \frac{4}{3} \Rightarrow \cos^2 \theta = \frac{2}{3}$$

$$\Rightarrow \cos \theta = \pm \frac{\sqrt{2}}{\sqrt{3}} \Rightarrow \tan \theta = \pm \frac{1}{\sqrt{2}}$$

$$\Rightarrow \tan^8 \theta = \left(\frac{1}{2^{1/2}} \right)^8 = \frac{1}{2^4} = \frac{1}{16}$$

$$\Rightarrow 32 \tan^8 \theta = 2 \cos^2 \alpha - 3 \cos \alpha$$

$$\Rightarrow 2 \cos^2 \alpha - 3 \cos \alpha = \frac{32}{16} = 2$$

$$\Rightarrow 2 \cos^2 \alpha - 3 \cos \alpha - 2 = 0$$

$$\Rightarrow 2 \cos^2 \alpha - 4 \cos \alpha + \cos \alpha - 2 = 0$$

$$\Rightarrow (2 \cos \alpha + 1)(\cos \alpha - 2) = 0 \Rightarrow \cos \alpha = -\frac{1}{2}, 2,$$

$$\because \cos \alpha \neq 2$$

$$\Rightarrow \cos \alpha = -\frac{1}{2} = \cos \left(\pi - \frac{\pi}{3} \right) = \cos \left(\frac{2\pi}{3} \right)$$

$$\Rightarrow \alpha = 2n\pi \pm \frac{2\pi}{3}$$



26. (a) $a \left(\frac{1+\cos C}{2} \right) + c \left(\frac{1+\cos A}{2} \right) = \frac{3b}{2}$

$$\therefore \cos^2 \frac{\theta}{2} = \frac{1+\cos \theta}{2}$$

$$\Rightarrow a + a \cos C + c + c \cos A = 3b$$

$$\Rightarrow a + c + (a \cos C + c \cos A) = 3b$$

Using projection formula,

$$a + c + b = 3b \Rightarrow 2b = a + c$$

Hence, the sides of triangle are in AP.

27. (a)

$$x^2 - 7xy + 12y^2 = 0$$

$$a = 1, h = -\frac{7}{2}, b = 12$$

$$\tan \theta = \frac{2\sqrt{h^2 - ab}}{|a+b|} \Rightarrow \tan \theta = \frac{2\sqrt{\frac{49}{4} - 12}}{13} \Rightarrow \tan \theta = \frac{1}{13}$$

$$\frac{2\cos \theta + 3\sin \theta}{4\sin \theta + 5\cos \theta} = \frac{2+3\tan \theta}{4\tan \theta + 5} = \frac{2+\frac{3}{13}}{\frac{4}{13}+5} = \frac{29}{69}$$

28. (b)

Let four geometric mean between 2 and 64 are

G_1, G_2, G_3, G_4 .

$\therefore 2, G_1, G_2, G_3, G_4, 64$ form a GP.

First term (a) = 2 and sixth term = 64

$ar^5 = 64, r$ = common ratio

$$2 \times r^5 = 64 \Rightarrow r^5 = 32 \Rightarrow r = 2$$

$$G_1 = ar = 2 \times 2 = 4$$

$$G_2 = ar^2 = 2(2)^2 = 8$$

$$G_3 = ar^3 = 2(2)^3 = 16$$

$$G_4 = ar^4 = 2(2)^4 = 32$$

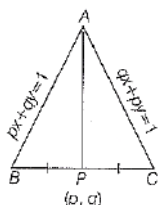
29. (d)

$$AB = px + qy = 1;$$

$$C = qx + py = 1$$

$$AP: AB + \lambda AC = 0$$

$$\text{i.e. } (px + qy - 1) + \lambda(qx + py - 1) = 0$$



Now Eq. (i) passes through (p, q)

$$\Rightarrow (p^2 + q^2 - 1) + \lambda(2pq - 1) = 0$$

$$\lambda = -\left(\frac{p^2 + q^2 - 1}{2pq - 1} \right)$$

$$\therefore AP: (px + qy - 1) - \left(\frac{p^2 + q^2 - 1}{2pq - 1} \right) (qx + py - 1) = 0$$

$$\Rightarrow (2pq - 1)(px + qy - 1) - (p^2 + q^2 - 1)$$

$$(qx + py - 1) = 0$$

30. (b) since $(a\hat{i} + \hat{j} + \hat{k})$, $(\hat{i} + b\hat{j} + \hat{k})$ and $(\hat{i} + \hat{j} + c\hat{k})$ are coplanar.

$$\therefore \begin{vmatrix} a & 1 & 1 \\ 1 & b & 1 \\ 1 & 1 & c \end{vmatrix} = 0$$

Apply $R_2 \rightarrow R_2 - R_1$ and $R_3 \rightarrow R_3 - R_1$

$$\begin{vmatrix} a & 1 & 1 \\ 1-a & b-1 & 0 \\ 1-a & 0 & c-1 \end{vmatrix} = 0$$

$$\Rightarrow (1-a)(1-b)(1-c) \begin{vmatrix} a & 1 & 1 \\ 1-a & b-1 & 0 \\ 1 & 0 & -1 \end{vmatrix} = 0$$

$$\Rightarrow \frac{a}{1-a} + \frac{1}{1-b} + \frac{1}{1-c} = 0 \Rightarrow \frac{a}{1-a} + 1 + \frac{1}{1-b} + \frac{1}{1-c} = 1$$

$$\Rightarrow \frac{1}{1-a} + \frac{1}{1-b} + \frac{1}{1-c} = 1$$

31. (d)

$$a = \hat{i} + \hat{j}, b = 2\hat{i} - \hat{k}$$

$$r \times a = b \times a \Rightarrow (r-b) \times a = 0 \Rightarrow r = b + \lambda a \dots (i)$$

$$\text{and } r \times b = a \times b \Rightarrow (r-a) \times b = 0 \Rightarrow r = a + \mu b \dots (ii)$$

where s, t are scalars

Eqs. (i) and (ii) are same if $s = t = 1$

$\therefore$ Point of intersection is $r = a + b$

$$\Rightarrow r = (\hat{i} + \hat{j}) + (2\hat{i} - \hat{k}) = 3\hat{i} + \hat{j} - \hat{k}$$

32. (a). $\Delta = 0$

$$\begin{vmatrix} 3 & -1 & 4 \\ 1 & 2 & -3 \\ 6 & 5 & \lambda \end{vmatrix} = 0$$

$$3(2\lambda + 15) + 1(\lambda + 18) + 4(5 - 12) = 0$$

$$\Rightarrow 6\lambda + 45 + \lambda + 18 + 20 - 48 = 0$$

$$\Rightarrow 7\lambda = -35 \Rightarrow \lambda = -5$$

33. (b)

$$\bar{x} = \frac{0 \times 1 + 1 \times {}^{50}C_1 + 2 \times {}^{50}C_2 + \dots + 50 \times {}^{50}C_{50}}{1 + {}^{50}C_1 + {}^{50}C_2 + \dots + {}^{50}C_{50}}$$

$$= \frac{\sum_{r=0}^{50} r \cdot {}^{50}C_r}{\sum_{r=0}^{50} {}^{50}C_r} = \frac{50 \cdot 2^{49}}{2^{50}} = \frac{50}{2} = 25$$

34. (d)

$$A \Delta B = (A \cup B) - (A \cap B) = (A - B) \cup (B - A)$$

$$A \cup B = \{1, 2, 3, 4, 5\}, A \cap B = \{3, 4\}$$

$$(A - B) = \{1, 2\}, (B - A) = \{5\}$$

$$(A - B) \cup (B - A) = A \Delta B = \{1, 2, 5\}$$

$$n(A \cup B) = 5, n(A \cap B) = 2, n(A \Delta B) = 3$$

$$5, n(A \cap B) = 2, n(A \Delta B) = 3 = 5 \times 2 \times 3 = 10 \times 3 = 30$$

35. (b)

$$n!(21-n)! = 21! \cdot \frac{n!(21-n)!}{21!} = 21! \cdot \frac{1}{{}^{21}C_n}$$

For minimum value of $n!(21-n)!$, ${}^{21}C_n$ is maximum

$$\text{Maximum value of } {}^{21}C_n = \frac{{}^{21}C_{21-1}}{2} = {}^{21}C_{10}$$

$$\therefore \text{Minimum value of } \frac{21!}{{}^{21}C_{10}} = \frac{21!}{21!} \times 10! \cdot 11!$$

Hence, $n = 10$.



36. (b)

Given that A_i 's are thirty sets with five element each.

$$\text{So } \sum_{i=1}^{30} n(A_i) = 5 \times 30 = 150 \quad \dots(i)$$

If there are m distinct elements in S and each element of S belongs to exactly 10 of the A_i 's, we have

$$\sum_{i=1}^{30} n(A_i) = 10m \quad \dots(ii)$$

From Eqs. (i) and (ii),

$$10m = 150 \Rightarrow m = 15$$

$$\text{Similarly } \sum_{j=1}^{30} n(B_j) = 3n \text{ and } \sum_{j=1}^{30} n(B_j) = 9m$$

$$\therefore 3n = 9m \Rightarrow n = \frac{9m}{3} = 3m = 3 \times 15 \Rightarrow n = 45$$

37. (c)

Given that, equation of ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$

Coordinate of any point on ellipse with eccentric angle θ are

$$(\alpha \cos \theta, b \sin \theta).$$

The coordinates of the end point of the latusrectum

$$= (ae, \pm \frac{b^2}{a})$$

$$\therefore a \cos \theta = ae \text{ and } b \sin \theta = \pm \frac{b^2}{a}$$

$$\Rightarrow \tan \theta = \pm \frac{b}{ae} \Rightarrow \theta = \tan^{-1} \left(\pm \frac{b}{ae} \right)$$

38. (d) $a^2 = 5\alpha - 3$ and $\beta^2 = 5\beta - 3$

$$\Rightarrow \alpha^2 - 5\alpha + 3 = 0 \text{ and } \beta^2 - 5\beta + 3 = 0$$

$\Rightarrow \alpha$ and β are roots of equation

$$\Rightarrow x^2 - 5x + 3 = 0$$

$$\Rightarrow \alpha + \beta = 5 \text{ and } \alpha\beta = 3$$

$$\Rightarrow \text{New Eq. has roots} = \frac{\alpha}{\beta} \text{ and } \frac{\beta}{\alpha}$$

$$\Rightarrow \text{Sum of roots} = \frac{\alpha}{\beta} + \frac{\beta}{\alpha} = \frac{\alpha^2 + \beta^2}{\alpha\beta} = \frac{(\alpha + \beta)^2 - 2\alpha\beta}{\alpha\beta}$$

$$= \frac{(5)^2 - 2(3)}{3} = \frac{25 - 6}{3} = \frac{19}{3}$$

$$\text{Product of roots} = \frac{\alpha}{\beta} \cdot \frac{\beta}{\alpha} = 1$$

$$\Rightarrow x^2 - (\text{Sum of roots})x + \text{Product of roots} = 0$$

$$\Rightarrow x^2 - \frac{19}{3}x + 1 = 0 \Rightarrow 3x^2 - 19x + 3 = 0$$

39. (d) Probability that a man aged x dies in a year = p

Therefore the probability that at least one man dies in a year = $1 - p$

No man dies in a year = $1 - (1 - p)^n$

Probability that out of n men A_1 is first to dies = $\frac{1}{n}$

Hence, the probability that A_1 dies in the one year

$$= \frac{1}{n} [1 - (1 - p)^n]$$

40. (d) $(a + xb) \cdot (a - b) = 0$

$$|a|^2 - a \cdot b + xa \cdot b - x|b|^2 = 0$$

$$|a|^2 + a \cdot b(x - 1) - 4x|a|^2 = 0$$

$$|a|^2(1 - 4x) + a \cdot b(x - 1) = 0$$

$$a \cdot b(x - 1) = (4x - 1)|a|^2$$

$$\Rightarrow |a||b|\cos 120^\circ(x - 1) = (4x - 1)|a|^2$$

$$\Rightarrow 2\left(\frac{-1}{2}\right)(x - 1) = (4x - 1) \Rightarrow 1 - x = 4x - 1 \Rightarrow 2 = 5x$$

$$\Rightarrow x = 2/5$$

41. (a) Simple space of selecting a number x from

$$1, 2, 3 \dots 100 = {}^{100}C_1 = 100$$

Solving the inequality $x + \frac{100}{x} > 29$

$$\frac{x^2 - 29x + 100}{x} > 0, \frac{(x - 25)(x - 4)}{x} > 0$$

$$0 < x < 4 \text{ Or } x > 25 \text{ Favourable cases} = [1, 2, 3, 26, \dots 100] = 78 \text{ number.}$$

$$\text{Probability} = \frac{78}{100} = \frac{39}{50}$$

42. (c)



$$X \cap \bar{Y} \cap (\bar{X} \cup \bar{Y}) = \phi$$

43. (b) $\int e^x (\sin hx + \cos hx) dx$

$$\sin hx = \frac{e^x - e^{-x}}{2}, \cos hx = \frac{e^x + e^{-x}}{2}, \frac{d}{dx} \cos hx = -\sin hx$$

Using formula,

$$\int e^x (f(x) + f'(x)) dx = e^x f(x) + C$$

$$\int e^x (\cos hx + \sin hx) dx = e^x \cos hx + C$$

44. (a)

$$e_1 = 2a + b, e_2 = a + 2b$$

$$\Rightarrow 2e_1 = 4a + 2b \Rightarrow e_2 = a + 2b$$

$$2e_1 - e_2 = 3a$$

$$\Rightarrow a = \frac{2e_1 - e_2}{3}$$

$$e_2 = \frac{2e_1 - e_2}{3} + 2b$$

$$\Rightarrow 2b = e_2 - \frac{2e_1 - e_2}{3}$$

$$= \frac{3e_2 - (2e_1 - e_2)}{3} = \frac{4e_2 - 2e_1}{3}$$

$$b = \frac{2e_2 - e_1}{3}$$

$$\text{Now, } a = \frac{2(\hat{i} + \hat{j} + \hat{k}) - (\hat{i} + \hat{j} - \hat{k})}{3} = \frac{\hat{i} + \hat{j} + 3\hat{k}}{3}$$

$$b = \frac{2(\hat{i} + \hat{j} - \hat{k}) - (\hat{i} + \hat{j} + \hat{k})}{3} = \frac{\hat{i} + \hat{j} - 3\hat{k}}{3}$$

$$a \cdot b = |a||b|\cos \theta$$



$$\Rightarrow \frac{(1+1-9)}{9} = \frac{\sqrt{11}}{3} \cdot \frac{\sqrt{11}}{3} \cos \theta$$

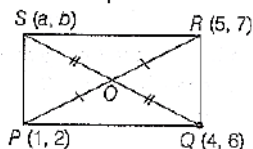
$$\Rightarrow -7 = 11 \cos \theta$$

$$\Rightarrow \cos \theta = \frac{-7}{11} \Rightarrow \theta = \cos^{-1}\left(\frac{-7}{11}\right)$$

45. (a)

In a parallelogram diagonals bisect each other.

O is mid point of PR and SQ.



$$O(x, y) = \left(3, \frac{9}{2}\right)$$

$$\therefore \left(3, \frac{9}{2}\right) = \left(\frac{a+4}{2}, \frac{b+6}{2}\right)$$

$$a = 2, b = 3$$

46. (a)

$$a \cos \theta + b \sin \theta = 2 \quad \dots\dots (i)$$

$$a \sin \theta - b \cos \theta = 3 \quad \dots\dots (ii)$$

Squaring and adding Eqs. (i) and (ii)

$$\Rightarrow a^2 + b^2 = 4 + 9 = 13$$

47. (d) Probability of getting triplet when 3 dice are thrown

$$= \frac{6}{6^3} = \frac{1}{36}$$

Probability of not getting triplet when 3 dice are thrown

$$= 1 - \left(\frac{1}{36}\right) = \frac{35}{36}$$

Here, we use binomial distribution.

Probability of throwing triplet not more than twice =

Probability (0 triplet) + Probability (1 triplet) +

Probability (2 triplets) = 1 - Probability of getting 3 triplets

$$\Rightarrow 1 - {}^3C_3 \left(\frac{1}{36}\right)^3 \Rightarrow 1 - \frac{1}{36^3}$$

48. (c) $\int 3^{3^x} 3^{3^x} 3^x dx$

$$\text{Put } 3^{3^x} = t \Rightarrow 3^{3^x} \log 3 \cdot 3^{3^x} \log 3 \cdot 3^x \log 3 \cdot dx = dt$$

$$3^{3^x} \cdot 3^{3^x} \cdot 3^x dx = \frac{dt}{(\log 3)^3}$$

$$= \int \frac{dt}{(\log 3)^3} = \frac{t}{(\log 3)^3} + C = \frac{3^{3^x}}{(\log 3)^3} + C$$

49. (a)

$${}^{50}C_1 + {}^{50}C_2 + {}^{50}C_2 + {}^{50}C_3 + \dots\dots {}^{50}C_{50}$$

$$\Rightarrow 2^{50} - 1$$

50. (b)

Class	Frequency	Cumulative Frequency
10 – 20	180	180
20 – 30	f_1	$180 + f_1$
30 – 40	34	$214 + f_1$
40 – 50	180	$394 + f_1$
50 – 60	136	$530 + f_1$
60 – 70	f_2	$530 + f_1 + f_2$
70 – 80	50	$580 + f_1 + f_2$

$$\text{Total frequency} = 686$$

$$580 + f_1 + f_2 = 686$$

$$\Rightarrow f_1 + f_2 = 106$$

$$\text{Median class} = \frac{N}{2} = \frac{686}{2} = 343$$

Median = 42.6 (given) lies in (40 – 50) interval Class 40 – 50

$$\text{Median} = l + \frac{\frac{N}{2} - C.F.}{f} \times h \text{ when } l = 40, h = 10$$

$$C.F = 214 + f_1$$

$$42.6 = 40 + \frac{\frac{686}{2} - (214 + f_1)}{180} \times 10$$

$$2.6 \times 18 = 343 - 214 - f_1$$

$$46.8 = 129 - f_1$$

$$\Rightarrow f_1 = 82.2 \approx 82$$

$$\text{Now } f_1 + f_2 = 106$$

$$\text{Put the value of } f_1$$

$$\Rightarrow f_2 = 106 - f_1$$

$$\Rightarrow f_2 = 106 - 82 = 24$$

REASONING

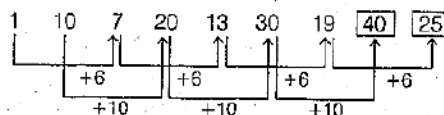
51. (b) VIBGYOR stands for Violet, Indigo, Blue, Green, Yellow, Orange and Red.

So, Yellow, Blue, Red and Orange are colours in VIBGYOR.

52. (b)

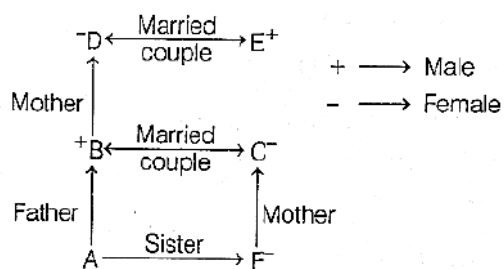
$$\begin{aligned} \text{Required percentage} &= \frac{48.17 - 37.16}{37.16} \times 100 \\ &= \frac{11.01}{37.16} = 29.62\% \approx 31\% \end{aligned}$$

53. (*) Pattern of the series as follows





54. (b) According to the given information



Hence DE is the one of couple.

55. (a) As, $Z = 52$

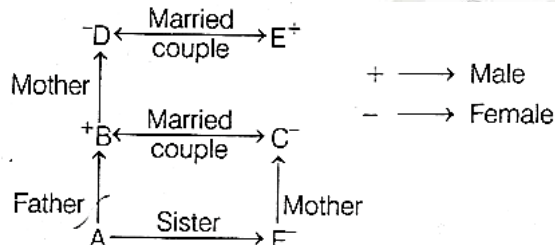
$$\Rightarrow 26 \times 2 = 52$$

and $ACT = 48$

$$\Rightarrow (1 + 3 + 20) \times 2 = 24 \times 2 = 48$$

$$\text{Similarly, } BAT \Rightarrow (2 + 1 + 20) \times 2 = 23 \times 2 = 46$$

Sol. (Q. Nos 56 and 57) According to given information,



56. (d) none of these statement are true.

57. (c) C is mother of A.

58. (d) "Trick" : Rope : Acrobat' is similar to 'Music : Guitar : performer'.

59. (b) As $9 \times 3 + 8 = 24$

$$\Rightarrow (9 \times 8) \div 3 = 72 \div 3 = 24; 10 \times 2 + 7 = 35$$

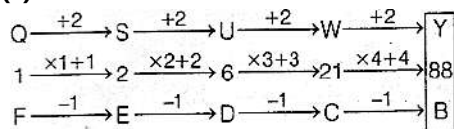
$$\Rightarrow (10 \times 7) \div 2 = 70 \div 2 = 35 \text{ and } 80 \times 40 + 3 = 6$$

$$\Rightarrow (80 \times 3) \div 40 = 240 \div 40 = 6$$

Similarly, $12 \times 4 + 3 = ?$

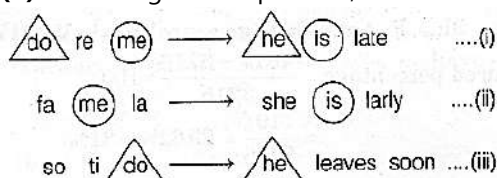
$$\Rightarrow (12 \times 3) \div 4 = 36 \div 4 = 9$$

60. (c) Pattern of series as follows



61. (b) The minimum qualifying marks in Chemistry for eligibility to symposium would lie in the range 20 – 30.

62. (d) According to the question,



From Eqs. (i), (ii) and (iii),
late = re

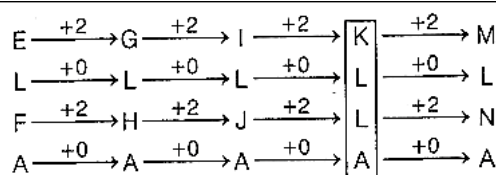
63. (c) As 7: 56

$$\Rightarrow 7: 7 \times (7 + 1) = 7: 7 \times 8 = 7: 56$$

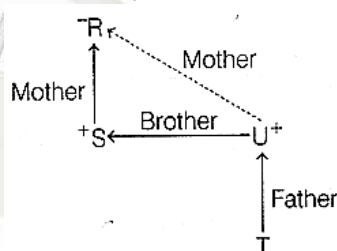
Similarly, 9: ?

$$\Rightarrow 9: 9 \times (9 + 1) = 9: 9 \times 10 = 9: 90$$

64. (d) Pattern of the series as follows



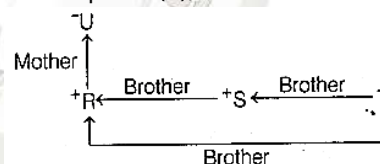
65. (b) from option A



$\therefore$ R is grandmother of T.

[incorrect]

From option (b),

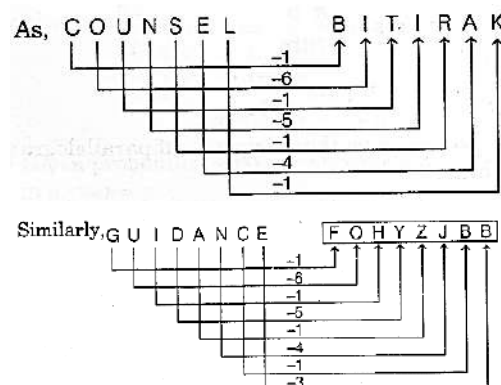


So, option (b) is correct.

66. (d) As, WAITRESS works in RESTAURANT.

Similarly, Teacher works in School.

67. (b)



68. (d) $15: 46 \Rightarrow 15 \times 3 + 1 = 46, 12: 37 \Rightarrow 12 \times 3 + 1 = 37,$

$$9: 28 \Rightarrow 9 \times 3 + 1 = 28 \text{ and } 8: 33 \Rightarrow 8 \times 3 + 1 = 25 \neq 33$$



69. (a) $\therefore$ Year 2000 is a leap year.

So, if it is Saturday on January 1, 2000, then January 1, 2021 would have been (Saturday + 2 = Monday).

70. (b) 2 yr ago, Rita's age = x (let) and Radha's present age = $2x$

According to the question, Difference between their ages = 2yr $\Rightarrow 2x - (x + 2) = 2 \Rightarrow 2x - x - 2 = 2 \Rightarrow x = 4$ Hence,

Radha's present age = $4 \times 2 = 8$ yr

71. (c) 22 times in a day are the hands of a clock pointing opposite to each other.

72. (b) Required marks for pursuing studies in Physics

$$= \frac{50 \times 60}{100} = 30$$

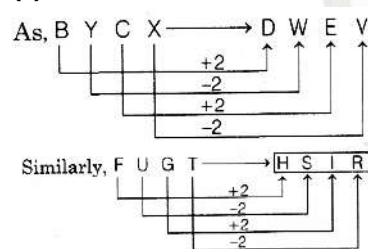
Hence, 32 students will be eligible to pursue higher studies in Physics.

73. (c) 40% of 50 = $50 \times \frac{40}{100} = 20$

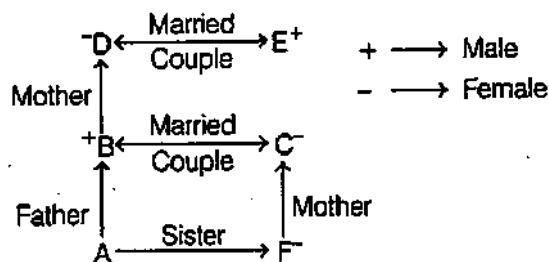
Hence, number of students scoring less than 40%.

Marks in aggregate = 100 - Number of students scoring 20 and above marks in aggregate = $100 - 73 = 27$

74. (c)



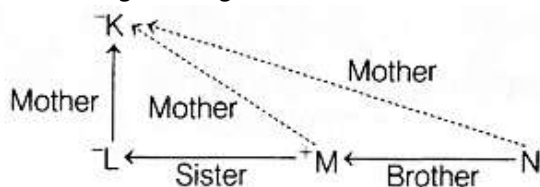
75. (c) According to the given information Gender of A cannot define



So number of male members are there in this family cannot be determine.

76. (a)

According to the given information



Hence K is the mother of N.

77. (a) Given,

$$X = 1! + 2! + 3! + \dots + 100!$$

$$\text{Now, } \frac{X}{24} = \left(\frac{1!+2!+3!+\dots+100!}{24} \right)$$

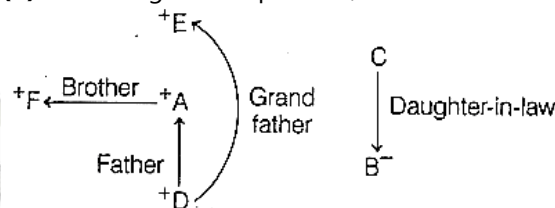
$$\Rightarrow \frac{X}{24} = \left(\frac{1 + 2 \times 1 + 3 \times 2 \times 1 + \dots + 100 \times 99 \times \dots \times 1}{24} \right)$$

$$\Rightarrow \frac{X}{24} = \left(\frac{1 + 2 + 6 + 24 + 120 + \dots + 100 \times 99 \times \dots \times 1}{24} \right)$$

$$\Rightarrow \frac{X}{24} = \frac{9}{24}$$

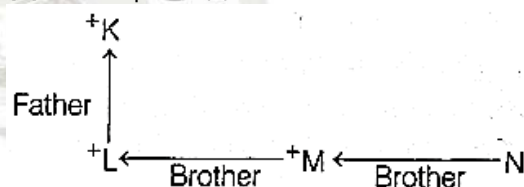
$\therefore$ Required remainder = 9

78. (d) According to the question,



Hence, relationship of C with F cannot defines. Hence, data inadequate.

79. (b) From option (a),



Hence, N is definitely daughter of K. (correct)

80. (b) Crop is a necessary part of Harvest.

81. (d) Required difference = Number of students scoring 30 and above marks in Chemistry - Number of students scoring 30 and above marks in aggregate = $27 - 21 = 6$

82. (c) Rate of increase for Product

$$A = \frac{48.17 - 20.15}{20.15} \times 100\% = 139.05\%$$

Rate of increase for Product

$$B = \frac{37.16 - 5.93}{5.93} \times 100\% = 526.64\%$$

Rate of increase for Product

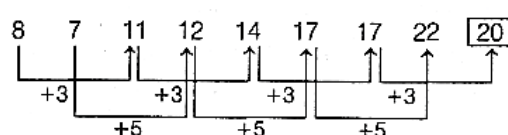
$$D = \frac{12.21 - 7.88}{7.88} \times 100\% = 54.94\%$$

Rate of increase for Product

$$E = \frac{10.19 - 5.01}{5.01} \times 100\% = 103.39\%$$

Hence, during the period 2005 - 2010, the minimum rate of increase in sales is in the case of product D.

83. (b) Pattern of the series as follows:





84. (c) As, Pulsate and Throb are synonyms.

Similarly, Examine and Scrutinize are synonyms.

85. (d) Total number of students in the class = 50

Let Paul's rank = x

So, Raghu's rank = $2x$

$\therefore$ 10 students who have ranked worse than that of Raghu.

$\therefore$ Raghu's rank = 40

So, Paul's rank = $\frac{40}{2} = 20$ th

86. (c) Except Hyena, all others are vegetarian.

87. (b)

As, A 3 T 15 R ACTOR

↑ ↑ ↑
Positional Value
Positional Value

and D 1 T 5 DATE

↑ ↑ ↑
Positional Value
Positional Value

Similarly, R 15 T 1 T 5 ROTATE

↑ ↑ ↑ ↑
Positional Value
Positional Value
Positional Value
Positional Value

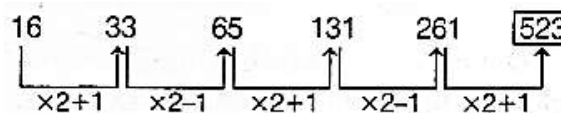
88. (d) Rate of increase for product

$$C = \frac{29.14 - 14.97}{14.97} \times 100 = 94.65\%$$

and from answer (82). It is clear, sales have increased by nearly 55% from 2005 to 2010 in case of product D.

89. (c) Except break, all others are parts of an airplane.

90. (a) Pattern of the series as follows



COMPUTER

91. A	92. D	93. D	94. B	95. A
96. C	97. a	98. b	99. d	100. a

ENGLISH

101. B	102. b	103. c	104. d	105. c
106. C	107. a	108. b	109. c	110. c
111. C	112. d	113. a	114. d	115. a
116. A	117. b	118. b	119. c	120. d



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